

Credit card transactions analysis

Project overview

This project focuses on analyzing the bank's credit card portfolio to gain insights into customer behavior, transaction performance, and credit risk. Using Excel, SQL, and Power BI, the data was cleaned, transformed, and visualized into an interactive dashboard. The report helps stakeholders track key business metrics, identify growth opportunities, and make informed decisions to enhance profitability and customer satisfaction.

Dataset Summery

- Table 1 : Credit_Card
- Rows : 10108
- Columns : 18
- Table 2 : Customer
- Rows : 10108
- Columns : 15

Data Loading & Transformation in SQL

Created table “cc_detail ” in sql

```
-- 1. Create cc_detail table

CREATE TABLE cc_detail (
    Client_Num INT,
    Card_Category VARCHAR(20),
    Annual_Fees INT,
    Activation_30_Days INT,
    Customer_Acq_Cost INT,
    Week_Start_Date VARCHAR(20),
    Week_Num VARCHAR(20),
    Qtr VARCHAR(10),
    current_year INT,
    Credit_Limit DECIMAL(10,2),
    Total_Revolving_Bal INT,
    Total_Trans_Amt INT,
    Total_Trans_Ct INT,
    Avg_Utilization_Ratio DECIMAL(10,3),
    Use_Chip VARCHAR(10),
    Exp_Type VARCHAR(50),
    Interest_Earned DECIMAL(10,3),
    Delinquent_Acc VARCHAR(5)
);
```

Created table “cust_detail ” in sql

```
-- 3. Create Customer details table
```

```
• CREATE TABLE cust_detail (  
    Client_Num INT,  
    Customer_Age INT,  
    Gender VARCHAR(5),  
    Dependent_Count INT,  
    Education_Level VARCHAR(50),  
    Marital_Status VARCHAR(20),  
    State_cd VARCHAR(50),  
    Zipcode VARCHAR(20),  
    Car_Owner VARCHAR(5),  
    House_Owner VARCHAR(5),  
    Personal_Loan VARCHAR(5),  
    Contact VARCHAR(50),  
    Customer_Job VARCHAR(50),  
    Income INT,  
    Cust_Satisfaction_Score INT  
);
```

Loaded CSV data into table “cc_details” from local system

```
-- 2. Load csv data into SQL
```

```
load data local infile 'D:/Data Analytics/E2E Project/Credit_Card_Dashboard/credit_card.csv'  
into table cc_detail  
fields terminated by ','  
enclosed by ''''  
lines terminated by '\n'  
ignore 1 rows;
```

Loaded CSV data into table “cust_details” from local system

```
-- 2. Load csv data into SQL
```

```
load data local infile 'D:/Data Analytics/E2E Project/Credit_Card_Dashboard/customer.csv'  
into table cust_detail  
fields terminated by ','  
enclosed by ''''  
lines terminated by '\n'  
ignore 1 rows;
```

Changed String to date format

```
-- . Changing str to date format

UPDATE cc_detail
SET Week_Start_Date = STR_TO_DATE(Week_Start_Date, '%d-%m-%Y');

ALTER TABLE cc_detail
MODIFY COLUMN Week_Start_Date DATE;
```

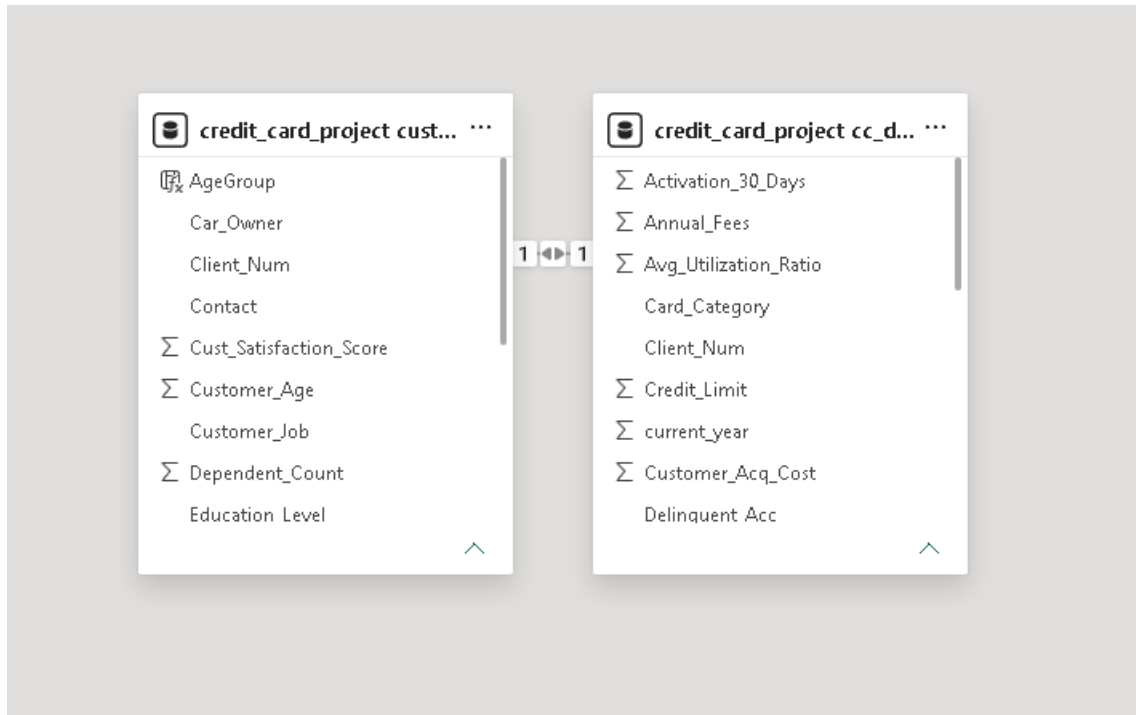
Checked data types before loading into Power BI

```
79
80 • desc cc_detail;
--
```

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
Field	Type	Null	Key	Default	Extra	
Client_Num	int	YES		NULL		
Card_Category	varchar(20)	YES		NULL		
Annual_Fees	int	YES		NULL		
Activation_30_Days	int	YES		NULL		
Customer_Acq_Cost	int	YES		NULL		
Week_Start_Date	date	YES		NULL		
Week_Num	varchar(20)	YES		NULL		
Qtr	varchar(10)	YES		NULL		
current_year	int	YES		NULL		
Credit_Limit	decimal(10,2)	YES		NULL		
Total_Revolving_Bal	int	YES		NULL		
Total_Trans_Amt	int	YES		NULL		
Total_Trans_Ct	int	YES		NULL		
Avg_Utilization_Ratio	decimal(10,3)	YES		NULL		
Use_Chip	varchar(10)	YES		NULL		
Exp_Type	varchar(50)	YES		NULL		
Interest_Earned	decimal(10,3)	YES		NULL		
Delinquent_Acc	varchar(5)	YES		NULL		

Insights & Visualizations in Power BI

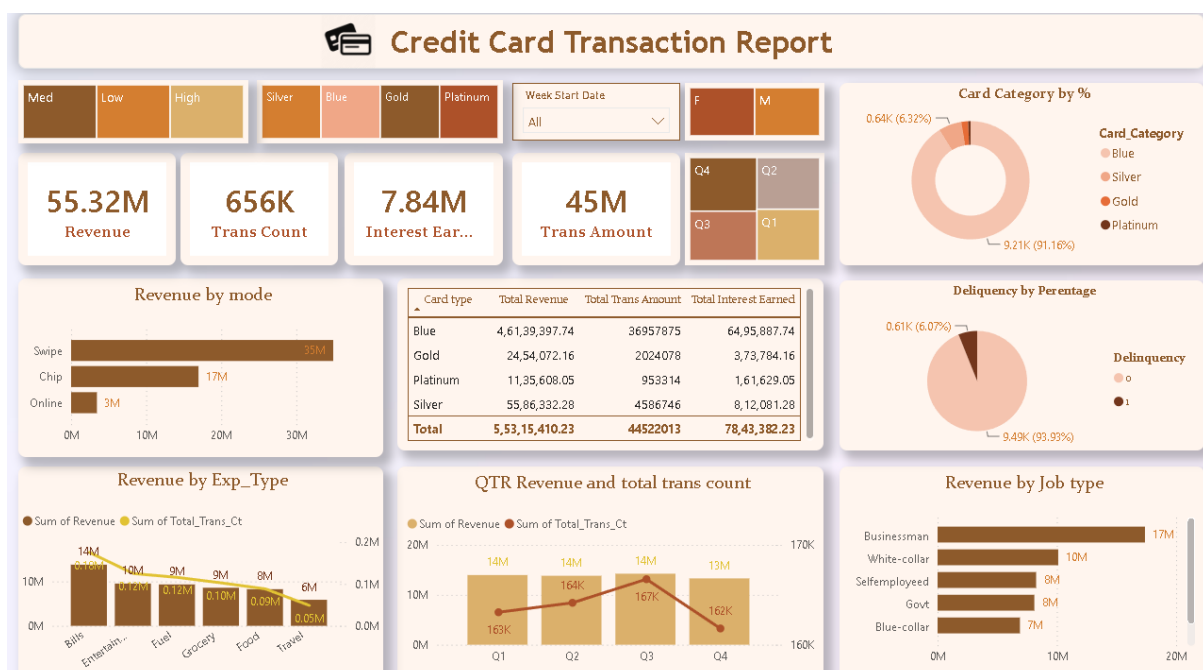
Data model in Power BI



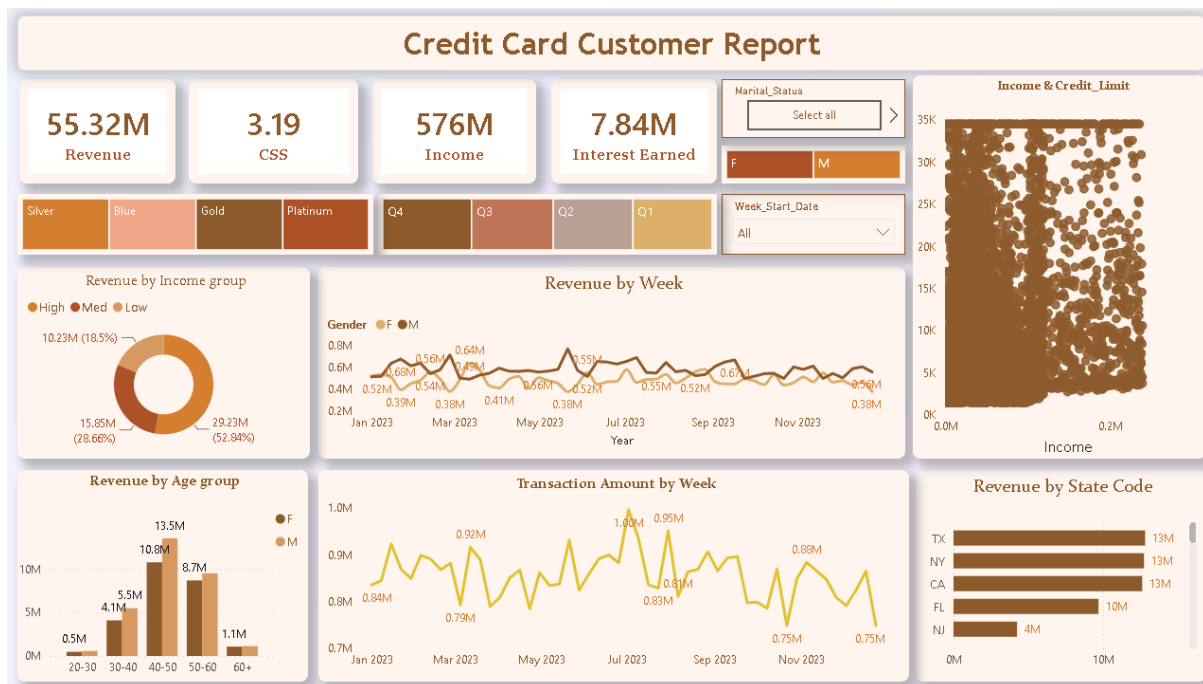
Performed required calculations using DAX queries and generate insights

Power BI Dashboard:

Credit Card Transactions Dashboard



Credit card customers report dashboard



Business recommendations

Upsell premium cards: Blue card holders form over **91%** of customers, while Gold and Platinum customers spend far more on average. Target high spending Blue customers for upgrades.

Increase card utilization: Identify customers with low utilization and launch cashback or rewards campaigns to encourage more spending.

Reduce credit risk: Build early warning monitoring for customers with high utilization and delinquent accounts, especially in high risk segments.

Improve customer satisfaction: Prioritize outreach to high-value customers with low satisfaction scores to reduce churn risk.

targeted marketing: Segment campaigns by income, occupation, age, and state rather than sending random offers.

Optimize credit limits: Regularly review credit limits for customers with strong repayment histories and high spending potential to maximize transaction volume.